



James Ruse Agricultural High School

2023 TASK 2 EXAMINATION

Mathematics Extension 2

General

Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/ or calculations

Total marks:
70

Section I – 5 marks (pages 2–3)

- Attempt Questions 1–5
- Allow about 8 minutes for this section

Section II – 65 marks (pages 4–8)

- Attempt Questions 6–10
- Allow about 1 hour and 52 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1 Given that $\frac{2x-1}{(x+1)(x-2)} = \frac{\alpha}{x+1} + \frac{\beta}{x-2}$, then

- A. $\alpha = -1$ and $\beta = 1$
- B. $\alpha = 1$ and $\beta = 1$
- C. $\alpha = -1$ and $\beta = -1$
- D. $\alpha = 1$ and $\beta = -1$

2 Which of the following definite integrals has the largest value?

- A. $\int_0^{\pi/2} \sin x \, dx$
- B. $\int_0^{\pi/2} \sin^2 x \, dx$
- C. $\int_0^{\pi/2} (1 - \sin x) \, dx$
- D. $\int_0^{\pi/2} (1 - \sin^2 x) \, dx$

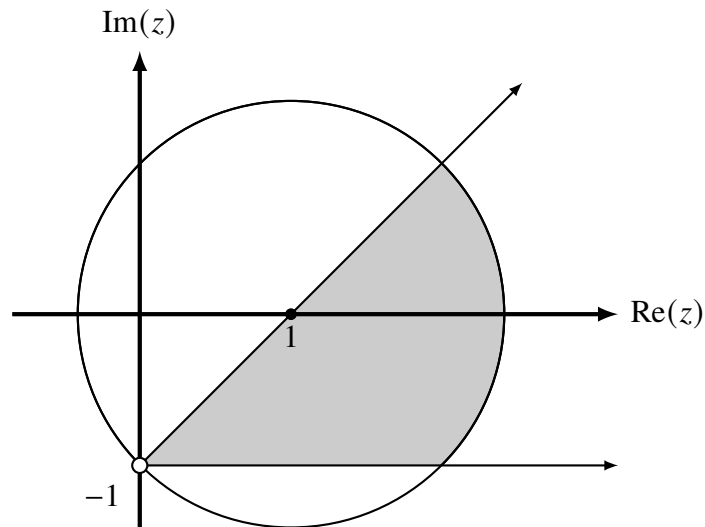
3 Christian thinks he has discovered a fact about even integers. He claims:

“Every even integer greater than 2 can be written as the sum of two prime numbers.”

Which of the following provides a counterexample to Christian’s claim?

- A. An even integer greater than 2 that can be written as the sum of two numbers which are not prime.
- B. An odd integer greater than 2 that can be written as the sum of two prime numbers.
- C. An even integer greater than 2 that cannot be written as the sum of two prime numbers.
- D. An odd integer greater than 2 that cannot be written as the sum of two prime numbers.

- 4 Consider the Argand diagram below.



Which inequality defines the shaded area?

- A. $|z - 1| \leq \sqrt{2}$ and $0 \leq \text{Arg}(z - i) \leq \frac{\pi}{4}$
B. $|z - 1| \leq \sqrt{2}$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$
C. $|z - 1| \leq 1$ and $0 \leq \text{Arg}(z - i) \leq \frac{\pi}{4}$
D. $|z - 1| \leq 1$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$

- 5 Consider the following two statements about two positive integers p and q :

P : The number $p^3 - q^3$ is even.

Q : The number $p + q$ is even.

Which of the following options is true?

- A. P is necessary and sufficient for Q .
B. P is necessary but not sufficient for Q .
C. P is sufficient but not necessary for Q .
D. P is neither necessary nor sufficient for Q .

Section II

65 marks

Attempt Questions 6–10

Allow about 1 hour and 52 minutes for this section

Write your answers on the paper supplied. Start each question on a new page.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks) Start this question on a new page

(a) Find the least positive integer k such that $\cos\left(\frac{4\pi}{7}\right) + i \sin\left(\frac{4\pi}{7}\right)$ is a solution of $z^k = 1$. **1**

(b) Let m and n be integers. Show that if the complex number w is a solution of $z^n = 1$, then so is w^m . **2**

(c) Find the following integrals:

(i) $\int \frac{1}{\sqrt{u} + 1} du$ **2**

(ii) $\int x \cos(x) \sin(x) dx$ **3**

(d) Consider the following statement:

P : “If I am tall and athletic, I am good at basketball and volleyball.”

(i) Write the contrapositive of P as an English sentence. **1**

(ii) Copy and complete the following sentence for the negation of P : **1**

“I am tall and athletic, ”

(e) Prove, by contrapositive, that if $\frac{mn}{2}$ is an integer, then m or n must be even. **3**

End of Question 6

Question 7 (11 marks) Start this question on a new page

(a) Let $z = \frac{3+4i}{5}$ and $w = \frac{5+12i}{13}$.

(i) Find zw and $z\bar{w}$, expressing your answer in $x + iy$ form. **2**

(ii) Hence, find two distinct ways of writing 65^2 in the form $a^2 + b^2$, where $a, b \in \mathbb{Z}$ and $0 < a < b < 65$. **1**

(b) By using a suitable substitution, or otherwise, evaluate **3**

$$\int_0^{\pi/2} \frac{d\theta}{2 + \cos \theta}.$$

(c) Suppose that b and c are odd integers. Show that the quadratic equation $x^2 - bx + c = 0$ cannot have repeated real roots. **3**

(d) Prove that there is only one pair of primes (p, q) such that $p^3 - q^3$ is also a prime number. **2**

End of Question 7

Question 8 (13 marks) Start this question on a new page

(a) Let 1, w and w^2 be the three cube roots of unity.

(i) Find the value of $(1 + 2w + 3w^2)(1 + 2w^2 + 3w)$. **2**

(ii) If the equations $x^3 - 1 = 0$ and $px^5 + qx + r = 0$ have a common root, evaluate **2**

$$(p + q + r)(pw^2 + qw + r)(pw + qw^2 + r).$$

(b) (i) Find $\int \frac{x}{1 + 2x + x^2} dx$. **2**

(ii) Hence, or otherwise, find the exact volume generated when the region bounded by the curve $y = \frac{e^x}{1 + e^x}$, the lines $x = -\ln 3$, $x = \ln 3$ and the x -axis is rotated about the x -axis. **3**

(c) (i) Find constants A and B such that **2**

$$4 \cos x - 8 \sin x = A(-2 \sin x - \cos x) + B(2 \cos x - \sin x).$$

(ii) Hence, find $\int \frac{4 \cos x - 8 \sin x}{2 \cos x - \sin x} dx$. **2**

End of Question 8

Question 9 (14 marks) Start this question on a new page

(a) Evaluate $\int_{\pi/6}^{\pi/2} \frac{\cos^3 x}{\sin^2 x} dx$. **3**

(b) (i) Show that $(1 - \sqrt{t})^{n-1} \sqrt{t} = (1 - \sqrt{t})^{n-1} - (1 - \sqrt{t})^n$. **1**

(ii) Define **4**

$$I_n = \int_0^1 (1 - \sqrt{t})^n dt \quad \text{for } t \geq 0.$$

Show that

$$I_n = \frac{n}{n+2} I_{n-1} \quad \text{for } n \geq 1.$$

(iii) Hence, show that $\int_0^1 (1 - \sqrt{t})^n dt = \frac{2n!}{(n+2)!}$. **2**

(c) Prove that the square of every prime number greater than 3 is one more than a multiple of 24. **4**

End of Question 9

Question 10 (14 marks) Start this question on a new page

(a) It is given that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$ (Do NOT prove this).

(i) Find the roots of $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$, expressing your answers in the form $x = \tan \alpha$. **3**

(ii) Hence, find the exact value of **1**

$$\tan^2 \left(\frac{\pi}{16} \right) + \tan^2 \left(\frac{3\pi}{16} \right) + \tan^2 \left(\frac{5\pi}{16} \right) + \tan^2 \left(\frac{7\pi}{16} \right).$$

(b) It is given that $w = r(\cos \theta + i \sin \theta)$ is a solution of the equation $z^n + az + 1 = 0$, where $\sin \theta \neq 0$ and $n > 2$ is an integer.

(i) Show that **1**

$$r^n \cos(n\theta) + ar \cos \theta + 1 = 0 \quad \text{and} \quad r^n \sin(n\theta) + ar \sin(\theta) = 0.$$

(ii) Hence, or otherwise, show that **2**

$$|w| \geq \sqrt[n]{\frac{1}{n-1}}.$$

You may assume that $|\sin(kx)| \leq k|\sin(x)|$ for all positive integers k .

(c) Suppose $f(x)$ is a function which satisfies the following: **4**

- $f''(x)$ is continuous
- $f(\pi) = 1$
- $\int_0^\pi (f(x) + f''(x)) \sin x \, dx = 2$

By using integration by parts, or otherwise, find $f(0)$.

(d) Prove that if n is a composite positive integer, then n divides $(n-1)!$ except in the case where $n = 4$. **3**

End of Question 10

End of Examination.

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Question 6:

- (a) Find the least positive integer k such that $\cos\left(\frac{4\pi}{7}\right) + i \sin\left(\frac{4\pi}{7}\right)$ is a solution of $z^k = 1$.

$$\left[\cos\left(\frac{4\pi}{7}\right) + i \sin\left(\frac{4\pi}{7}\right)\right]^k = 1$$

$$\therefore \cos\frac{4k\pi}{7} + i \sin\frac{4k\pi}{7} = 1$$

$$\therefore \frac{4k\pi}{7} = n\pi, n \in \mathbb{Z}$$

$$\therefore k = \frac{7n}{4}, \text{ so } n = 4 \text{ and } \boxed{k = 7}.$$

1 mark

- (b) Let m and n be integers. Show that if the complex number w is a solution of $z^n = 1$, then so is w^m .

$$\text{suppose } w \text{ is a solution of } z^n = 1 \therefore w^n = 1$$

$$(w^m)^n = (w^n)^m = 1^m = 1$$

$$\therefore w^m \text{ is a solution of } z^n = 1.$$

1 mark for sub w^m
into z^n .

1 mark for showing
that $(w^m)^n = 1$

- (c) Find the following Integrals:

(i) $\int \frac{1}{\sqrt{u}+1} du$

$$\text{Let } x = \sqrt{u} + 1$$

$$dx = \frac{1}{2\sqrt{u}} du$$

$$du = 2\sqrt{u} dx$$

$$= 2(x-1) dx$$

$$\int \frac{1}{\sqrt{u}+1} du = \int \frac{2(x-1)}{x} dx = 2 \int \left(1 - \frac{1}{x}\right) dx \quad \checkmark$$

$$= 2x - 2 \ln|x| + C$$

$$= 2(\sqrt{u}+1) - 2 \ln(\sqrt{u}+1) + C \quad \checkmark$$

Note: Final answer must be in terms of u .

$$(ii) \quad \int x \cos(x) \sin(x) dx = \frac{1}{2} \int x \sin 2x dx$$

$$\text{let } u = x \quad \text{and} \quad dv = \sin 2x dx$$

$$du = dx \quad \text{and} \quad v = -\frac{1}{2} \cos 2x$$

$$\int x \cos x \sin x dx = \frac{1}{2} \left[-\frac{1}{2} x \cos 2x - \int \left(-\frac{1}{2} \cos 2x \right) dx \right]$$

$$= -\frac{1}{4} x \cos 2x + \frac{1}{4} \int \cos 2x dx$$

$$= -\frac{1}{4} x \cos 2x + \frac{1}{8} \sin 2x + C$$

Note: you can not write $dv = \sin 2x$ without dx

(d) Consider the following statement:

P: "If I am tall and athletic, I am good at basketball and volleyball."

(i) Write the contrapositive of P as an English Sentence.

If I am not good at either basketball or volleyball
then I am either not tall or not athletic

(ii) Copy and complete the following sentence for the negation of P.

"I am tall and athletic, ~~and~~ I am bad at either basketball or volleyball."

Note: $P \Rightarrow Q \equiv \neg(P \wedge \neg Q)$
 $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$

(Q) Prove, by contrapositive, that if $\frac{mn}{2}$ is an integer, then m and n must be even.

The Contrapositive is:

If m and n are odd, then $\frac{mn}{2}$ is not an integer.

Suppose $m = 2k+1$ and $n = 2j+1$, $k, j \in \mathbb{Z}$

$$\begin{aligned}\frac{mn}{2} &= \frac{(2k+1)(2j+1)}{2} \\ &= \frac{4kj + 2(k+j) + 1}{2} \\ &= 2kj + k + j + \frac{1}{2} \\ &= M + \frac{1}{2}, \text{ where } M = 2kj + k + j \in \mathbb{Z}\end{aligned}$$

but $\frac{1}{2} \notin \mathbb{Z} \therefore M + \frac{1}{2} \notin \mathbb{Z} \therefore \frac{mn}{2} \notin \mathbb{Z}$.

Note: you need to justify your answer by saying clearly that $\frac{1}{2}$ is not an integer.

Note: you need to write the contrapositive statement

Question 7

a)

$$\begin{aligned}\text{i. } zw &= \left(\frac{3+4i}{5}\right)\left(\frac{5+12i}{13}\right) \\ &= \frac{(3+4i)(5+12i)}{65} \\ &= \frac{15+36i+20i-48}{65} \\ &= \frac{-33+56i}{65} \\ &= -\frac{33}{65} + \frac{56i}{65}\end{aligned}$$

1 mark

$$\begin{aligned}z\bar{w} &= \left(\frac{3+4i}{5}\right)\left(\frac{5-12i}{13}\right) \\ &= \frac{(3+4i)(5-12i)}{65} \\ &= \frac{15-36i+20i+48}{65} \\ &= \frac{63-16i}{65} \\ &= \frac{63}{65} - \frac{16i}{65}\end{aligned}$$

1 mark

$$\begin{aligned}\text{ii. } |z| &= |\bar{z}| = |w| = |\bar{w}| = 1 \\ \therefore |zw| &= 1 \rightarrow \frac{1}{65} |-33+56i| = 1 \\ \therefore \sqrt{(-33)^2 + 56^2} &= 65 \\ \therefore 33^2 + 56^2 &= 65^2\end{aligned}$$

$$\begin{aligned}\text{Similarly, } |z\bar{w}| &= 1 \rightarrow \frac{1}{65} |63-16i| = 1 \\ \therefore \sqrt{(63)^2 + (-16)^2} &= 65 \\ \therefore 63^2 + 16^2 &= 65^2\end{aligned}$$

1 mark for answer that shows the acknowledgement of $|zw| = 1$ and $|z\bar{w}| = 1$

However, the sums MUST be made up of positive integers

b)

$$\text{Let } I = \int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$$

$$\text{Let } t = \tan\left(\frac{\theta}{2}\right)$$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2\left(\frac{\theta}{2}\right)$$

$$\frac{d\theta}{dt} = 2 \cos^2\left(\frac{\theta}{2}\right)$$

$$d\theta = \frac{2}{1+t^2} dt$$

$$\text{When } \theta = \frac{\pi}{2}, t = 1$$

$$\theta = 0, t = 0$$

$$\therefore I = \int_0^1 \frac{\frac{2}{1+t^2}}{2 + \left(\frac{1-t^2}{1+t^2}\right)} dt$$

$$= \int_0^1 \frac{2}{t^2 + 3} dt$$

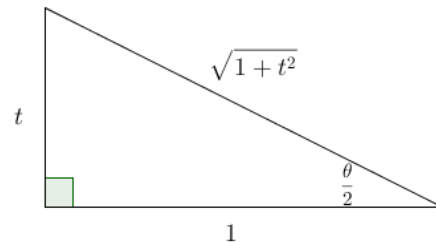
$$= \frac{2}{\sqrt{3}} \int_0^1 \frac{\sqrt{3}}{t^2 + 3} dt$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1}\left(\frac{t}{\sqrt{3}}\right) \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

$$= \frac{2}{\sqrt{3}} \times \frac{\pi}{6}$$

$$= \frac{\pi}{3\sqrt{3}}$$



1 mark for finding ALL necessary information for the substitution.

1 mark for correctly integrating into an inverse tan function.

1 mark for answer.

c)

Assume that the quadratic equation $x^2 - bx + c = 0$ has repeated real roots, and

$b = 2m + 1$ and $c = 2n + 1, m, n \in \mathbb{Z}$.

$$\Delta = 0 \rightarrow (-b)^2 - 4(1)(c) = 0$$

1 mark for recognising $\Delta = 0$ in the assumption

$$b^2 - 4c = 0$$

$$(2m + 1)^2 - 4(2n + 1) = 0$$

1 mark for substitution odd expressions into discriminant

$$\therefore (2m + 1)^2 = 4(2n + 1)$$

Which is a contradiction since:

- LHS is an odd number since the square of an odd number must be odd.

- RHS is an even number since it is divisible by 4, hence divisible by 2.

$\therefore x^2 - bx + c = 0$ cannot have real repeated roots if b and c are both odd.

1 mark for arriving to a contradiction and conclusion.

Alternatively,

Let $f(x) = x^2 - bx + c$ and the repeated root for $f(x) = 0$ be α .

Since $f(x)$ is monic and b and c are integers, α must be an integer.

1 mark

Repeated root $\rightarrow f'(\alpha) = 0$

1 mark

$$\therefore 2\alpha - b = 0$$

$$b = 2\alpha$$

$\therefore b$ is even, which is a contradiction as b is defined to be odd.

$\therefore x^2 - bx + c = 0$ cannot have real repeated roots if b and c are both odd. 1 mark

Student MUST explain why the root of the equation has to be an integer before using that as a premise in your working.

d)

$$p^3 - q^3 = (p - q)(p^2 + pq + q^2)$$

$$p^3 - q^3 \text{ is prime} \rightarrow (p - q = 1 \text{ and } p^2 + pq + q^2 > 1) \text{ or } (p - q > 1 \text{ and } p^2 + pq + q^2 = 1)$$

$$p \text{ and } q \text{ are prime} \rightarrow p^2 + pq + q^2 \neq 1 \text{ as } p, q \geq 2$$

$$\therefore p^2 + pq + q^2 > 1 \text{ and } p - q = 1$$

Since there are only 2 prime numbers that have a difference of 1, namely 2 and 3, then, there is only one pair of prime (p, q) such that $p^3 - q^3$ is also a prime.

1 mark for getting the implication for if $p^3 - q^3$ is prime.

$$\text{i.e. } (p - q = 1 \text{ or } p^2 + pq + q^2 = 1)$$

1 mark for final answer.

MATHEMATICS EXT 2: Question 8

Suggested Solutions	Marks	Marker's Comments
a.i. Note that $1+\omega+\omega^2=0$ & $\omega^3=1$. $(1+2\omega+3\omega^2)(1+2\omega^2+3\omega)$ $= 1+2\omega^2+3\omega+2\omega+4\omega^3+6\omega^2+3\omega^2+6\omega^4+9\omega^3$ $= 1+5\omega+11\omega^2+13\omega^3+6\omega^4$ $= 14+11\omega+11\omega^2, \text{ since } \omega^3=1$ $= 3+11(1+\omega+\omega^2)$ $= 3+11 \times 0$ $= 3$	① ①	Using $\omega^3=1$ with reasonable progress. Final answer
ii. Since $x^3-1=0$ & $px^5+qx+r=0$ share a common root, the root must be either 1, ω or ω^2. <u>Case 1:</u> 1 is the common root Then $p(1)^5+q(1)+r=0$ $p+q+r=0$ $\Rightarrow (p+q+r)(p\omega^2+q\omega+r)(p\omega+\omega^2+r)=0$ <u>Case 2:</u> ω is the common root Then $p\omega^5+q\omega+r=0$ $p\omega^2+q\omega+r=0, \text{ since } \omega^3=1$ $\Rightarrow (p+q+r)(p\omega^2+q\omega+r)(p\omega+\omega^2+r)=0$ <u>Case 3:</u> ω^2 is the common root Then $p\omega^{10}+q\omega^2+r=0$ $p\omega+q\omega^2+r=0, \text{ since } \omega^3=1$ $\Rightarrow (p+q+r)(p\omega^2+q\omega+r)(p\omega+\omega^2+r)=0$ In either case, the product is always equal to 0.	① ① ①	Fully correct working. Most students falsely claimed that $z=1$ was THE root, or that $z=\omega$ was THE root. This reduces the generality and thus is an incomplete solution.

MATHEMATICS EXT 2: Question 8

Suggested Solutions	Marks	Marker's Comments
bi. $\int \frac{x}{1+2x+x^2} dx$ $= \int \frac{x}{(1+x)^2} dx$ $= \int \left(\frac{x+1}{(1+x)^2} - \frac{1}{(1+x)^2} \right) dx$ $= \int \left(\frac{1}{1+x} - (1+x)^{-2} \right) dx$ $= \ln 1+x + \frac{1}{1+x} + C$	① ①	Algebraic manipulation Final answer $\Rightarrow$ MUST have absolute values.
ii. $V = \pi \int_{-\ln 3}^{\ln 3} \left(\frac{e^x}{1+e^x} \right)^2 dx$ $= \pi \int_{-\ln 3}^{\ln 3} \frac{e^x \cdot e^x}{1+2e^x+e^{2x}} dx$ $= \pi \int_{1/3}^3 \frac{u}{1+2u+u^2} du$ $= \pi \left[\ln 1+u + \frac{1}{1+u} \right]_{1/3}^3$ $= \pi \left(\ln 4 + \frac{1}{4} - \left(\ln \left(\frac{4}{3} \right) + \frac{3}{4} \right) \right) u^3$ $= \pi \left(\ln 3 - \frac{1}{2} \right) u^3$	let $u=e^x$ $du=e^x dx$ $x=\ln 3 \rightarrow u=3$ $x=-\ln 3 \rightarrow u=\frac{1}{3}$ ① ① ①	Recognising $u=e^x$ sub or equivalent Using part (i) Final answer

MATHEMATICS EXT 2: Question 8

Suggested Solutions	Marks	Marker's Comments
ci, $4\cos x - 8\sin x = (2B - A)\cos x + (-2A - B)\sin x$ $\Rightarrow \begin{cases} 4 = 2B - A \\ -8 = -2A - B \end{cases}$ Solving simultaneously: $A = \frac{12}{5} \quad B = \frac{16}{5}$		
ii, $\int \frac{4\cos x - 8\sin x}{2\cos x - \sin x} dx$ $= \int \frac{\frac{12}{5}(-2\sin x - \cos x) + \frac{16}{5}(2\cos x - \sin x)}{2\cos x - \sin x} dx$ $= \frac{12}{5} \ln 2\cos x - \sin x + \frac{16}{5} x + C$	(2) (1) (1)	One for each. Using part (i) Final answer.

9

MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
a) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin^2 x} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x (1 - \sin^2 x)}{\sin^2 x} dx$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin^2 x} - \cos x dx$ $= \left[\frac{-1}{\sin x} - \sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$ $= \frac{-1}{\sin \frac{\pi}{2}} - \sin \frac{\pi}{2} - \left(\frac{-1}{\sin \frac{\pi}{6}} - \sin \frac{\pi}{6} \right)$ $= -1 - 1 - (-2 - \frac{1}{2})$ $= \frac{1}{2}$	① ① ①	
b) i. RHS = $(1 - \sqrt{t})^{n-1} - (1 - \sqrt{t})^n$ $= (1 - \sqrt{t})^{n-1} [1 - (1 - \sqrt{t})]$ $= (1 - \sqrt{t})^{n-1} \sqrt{t} = \text{LHS}$	①	
ii. $I_n = \int_0^1 (1 - \sqrt{t})^n dt$ $= [t(1 - \sqrt{t})^n]_0^1 - \int_0^1 t \times \frac{-n}{2\sqrt{t}} (1 - \sqrt{t})^{n-1} dt$ $= (1(1-1)^n - 0) + \frac{n}{2} \int_0^1 \sqrt{t} (1 - \sqrt{t})^{n-1} dt$ $= \frac{n}{2} \int_0^1 (1 - \sqrt{t})^{n-1} = (1 - \sqrt{t})^n dt$ $= \frac{n}{2} (I_{n-1} - I_n)$ $2I_n = n I_{n-1} - n I_n$ $I_n (2+n) = n I_{n-1}$ $I_n = \frac{n}{2+n} I_{n-1}$	① ① ① ①	Recognising by parts Applying by parts correctly Many students used part i) too early on and got a statement of $I_n = I_n$ or $I_{n-1} = I_{n-1}$. No marks awarded for that.
iii) $I_n = \frac{n}{n+2} I_{n-1}$ $= \frac{n}{n+2} \times \frac{n-1}{n+1} I_{n-2}$ $= \frac{n}{n+2} \times \frac{n-1}{n+1} \times \frac{n-2}{n} I_{n-3}$	①	

MATHEMATICS Extension 2: Question.....9

Suggested Solutions	Marks	Marker's Comments
$= \frac{n}{n+2} \times \frac{n-1}{n+1} \times \frac{n-2}{n} \times \dots \times \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} \times I_0$ $I_0 = \int_0^1 dt$ $= 1$ $\therefore I_n = \frac{n!}{(n+2)(n+1)n \times \dots \times 4 \times 3} \times \frac{2}{2} \times 1$ $= \frac{2n!}{(n+2)!}$		Could use I_1 as ending of the reduction.
c) let $p > 3$ be prime. p must be odd (only even prime is 2) prove $p^2 = 24N + 1$, $N \in \mathbb{Z}$ Consider: $p^2 - 1 = (2k+1)^2 - 1$ $= 4k^2 + 4k + 1 - 1$ $= 4k(k+1)$ Since k and $(k+1)$ are consecutive, one must be even. i.e. $k(k+1) = 2M$, $M \in \mathbb{Z}$ $\therefore p^2 - 1 = 4(2M)$ $= 8M$ $(p-1) \times p \times (p+1) = 3N$, $N \in \mathbb{Z}$ as they are 3 consecutive numbers Since p is prime the factor of 3 lies in $(p-1) \times (p+1)$	① ① ① ①	Had to show I_0 or I_1 and then adjustment for factorial in denominator.
$\therefore p^2 - 1 = 8 \times 3N$, $N \in \mathbb{Z}$ as 3 and 8 are coprime $\therefore p^2 - 1 = 24N$ $p^2 = 24N + 1$	①	
OR Possible primes occur at $6k+1$ or $6k-1$ (remove all numbers with factors of 2 and/or 3) for some integer k CASE 1: $(6k+1)^2 = 36k^2 + 12k + 1$ $= 12(3k+1)k + 1$ If k is odd then $3k$ is odd and $3k+1$ is even. If k is even then $3k$ is even and $3k+1$ is odd i.e. $k(3k+1) = 2M$, $M \in \mathbb{Z}$ $\therefore \text{Our prime can be expressed as } 12(2M) + 1$	①	can also use $6k+5$

MATHEMATICS Extension 2: Question...9....

Suggested Solutions	Marks	Marker's Comments
or $24M + 1$ i.e. ^{one greater than} a multiple of 24 CASE 2: $(6k-1)^2 = 36k^2 - 12k + 1$ $= 12(3k-1)k + 1$ If k is odd then $3k$ is odd and $3k-1$ is even. If k is even then $3k$ is even and $3k-1$ is odd i.e. $k(3k-1) = 2N, N \in \mathbb{Z}$ $\therefore$ Our ^{squared} prime can be expressed as $12(2N) + 1$ or $24N + 1$ i.e. one greater than a multiple of 24.	①	THIS CANNOT BE DONE BY MATHEMATICAL INDUCTION.

Question 10

a, i, $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$

$$x^4 - 6x^2 + 1 = 4x - 4x^3$$

$$1 = \frac{4x - 4x^3}{x^4 - 6x^2 + 1}$$

let $x = \tan \theta$

$$1 = \frac{4 \tan \theta - 4 \tan^3 \theta}{\tan^4 \theta - 6 \tan^2 \theta + 1}$$

$$1 = \tan 4\theta \quad (\text{given})$$

$$4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}, \dots$$

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}, \dots$$

$\therefore$ Roots are $x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$

$$x = \tan\left(\frac{\pi}{16} + \frac{n\pi}{4}\right) : \text{max } 2$$

ii, $\tan \frac{9\pi}{16} = \tan\left(-\frac{7\pi}{16}\right)$
 $= -\tan \frac{7\pi}{16}$

$$\tan \frac{13\pi}{16} = \tan\left(-\frac{3\pi}{16}\right)$$
$$= -\tan \frac{3\pi}{16}$$

$\therefore$ The four roots of equation in part i, can be written as

$$x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{3\pi}{16}, -\tan \frac{7\pi}{16}$$

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16}$$

$$= \tan^2 \frac{\pi}{16} + \left(-\tan \frac{3\pi}{16}\right)^2 + \tan^2 \frac{5\pi}{16} + \left(-\tan \frac{7\pi}{16}\right)^2$$

$$= (\sum \alpha)^2 - 2(\sum \alpha\beta)$$

$$= (-4)^2 - 2(-6)$$

$$= 16 + 12$$

$$= 28$$

✓ CNE allow

✓
'Hence' Q
show the link

no mark for
answer only

b, i, $w = r(\cos\theta + i\sin\theta)$ is a solⁿ of $z^n + az + 1 = 0$

$$\therefore (r \operatorname{cis} \theta)^n + a(r \operatorname{cis} \theta) + 1 = 0 \quad \text{for } \operatorname{cis} \theta = \cos \theta + i \sin \theta$$

allow cis

$$r^n \operatorname{cis}(n\theta) + ar \operatorname{cis} \theta + 1 = 0$$



Equating real part: $r^n \cos(n\theta) + ar \cos \theta + 1 = 0$

Equating imaginary part: $r^n \sin(n\theta) + ar \sin \theta = 0$

⊖ Equating coefficients

⊖ Equating complex part

c, $\int_0^\pi (f(x) + f''(x)) \sin x \, dx$

$$= \int_0^\pi f(x) \sin x \, dx + \int_0^\pi f''(x) \sin x \, dx$$

$$\begin{matrix} u' & v \\ u = f'(x) & v' = \cos x \end{matrix}$$

⊖ $\int_0^\pi (f(x) + f''(x)) \sin x \, dx$
u v'

$$= \int_0^\pi f(x) \sin x \, dx + \left[f'(x) \sin x \right]_0^\pi - \int_0^\pi f'(x) \cos x \, dx \quad \checkmark \text{ by parts}$$

$$= \int_0^\pi f(x) \sin x \, dx + \underline{0} - \int_0^\pi f'(x) \cos x \, dx \quad \checkmark \text{ eval.}$$

$$\begin{matrix} u' & v \\ u = f(x) & v' = -\sin x \end{matrix}$$

$$= \int_0^\pi f(x) \sin x \, dx - \left[f(x) \cos x \right]_0^\pi - \int_0^\pi f(x) \sin x \, dx \quad \checkmark \text{ by parts}$$

$$= \left[f(x) \cos x \right]_\pi^0$$

$$= f(0) \cos(0) - f(\pi) \cos(\pi)$$

$$= f(0) + 1$$

as $f(\pi) = 1$ given

Since $\int_0^\pi (f(x) + f''(x)) \sin x \, dx = 2$ given,

$$f(0) + 1 = 2$$

$$\therefore f(0) = 1$$



d, Suppose $n = ab$, where $a, b \in \mathbb{Z}$ *

$$\underline{1 < a \leq b < n}$$

↳ or $2 \leq a \leq b \leq n-1$

Case I if $a \neq b$

Since a and b are distinct integers less than n , both a and b show up as different factors in $(n-1)!$

$$\therefore n = ab \mid (n-1)!$$



Case II if $a = b = 2$

$$n = 4 \nmid 3! = 6$$

$$\therefore n \nmid (n-1)! \quad \text{for } n=4$$



Case III if $a = b > 2$

$$\text{i.e. } n = a^2 > 4$$

$$\text{For } a > 2, \quad 2a < a^2$$

$$2a \leq a^2 - 1$$

$$2a \leq n - 1$$

$$\therefore 2a \mid (n-1)!$$

But, also $a \mid (n-1)!$ as $a \in \mathbb{Z}$, $a < n$

Since a and $2a$ are distinct integers less than n

$$2a^2 \mid (n-1)!$$

$$\therefore n = a^2 \mid (n-1)!$$



$\therefore$ If n is a composite positive integer, then $n \mid (n-1)!$

except when $n = 4$.